

Lecture 3: The Web Trio – HTML, CSS, and JavaScript

Goal: Understand how the three core technologies work together to create every website you visit.

1. HTML: The Skeleton (Structure)

HTML stands for **HyperText Markup Language**. It is not a programming language; it is a **marking** language. It tells the browser what each part of your page *is*.

- **Real-life Example:** Think of HTML like the **blueprint of a house**. It defines where the walls go, where the doors are placed, and how many rooms there are. It doesn't tell you the color of the paint or the type of furniture; it just defines the **structure**.
- **Key Concept:** HTML uses **tags**.
 - `<h1>` for a main title.
 - `<p>` for a paragraph of text.
 - `<img>` for an image.
- **Why it matters:** Without HTML, your website is just a jumbled mess of text with no meaning or order.

2. CSS: The Style (Presentation)

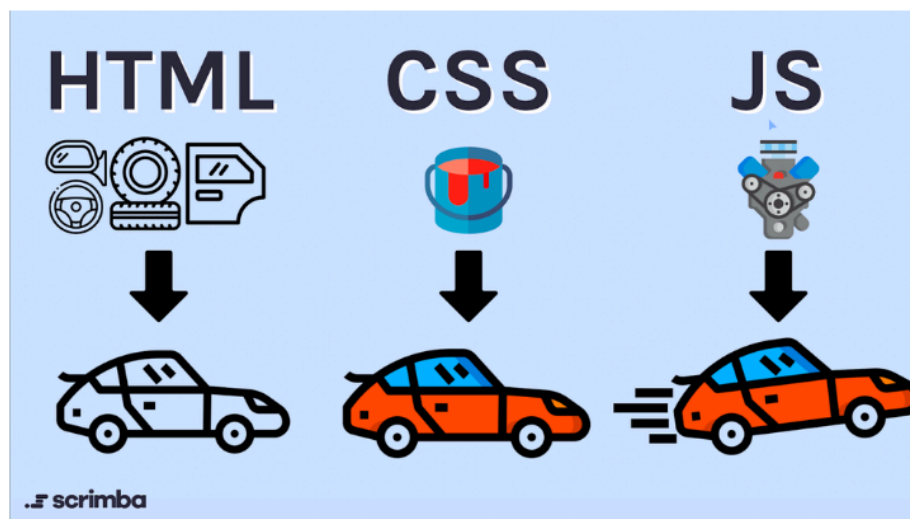
CSS stands for **Cascading Style Sheets**. This is what makes your website look professional, modern, and branded.

- **Real-life Example:** If HTML is the house blueprint, CSS is the **interior design**. It decides:
 - What color are the walls? (Background colors)
 - How big are the windows? (Width and height)
 - Where do the pictures hang? (Layout and positioning)
- **Key Concept:** CSS uses **rules** to target HTML elements.
 - "Change all `<h1>` titles to blue."
 - "Make all paragraphs 16 pixels large."
- **Why it matters:** CSS is what turns a boring document into a beautiful, user-friendly interface. It allows for **Responsiveness**—ensuring the site looks good on both a giant monitor and a small mobile phone.

3. JavaScript: The Brain (Behavior)

JavaScript (JS) is the only true programming language of the three. It makes the page interactive.

- **Real-life Example:** If HTML is the house and CSS is the decoration, JavaScript is the **electricity and smart-home system**.
 - Click a button to turn on a light? That's JS.
 - An alarm goes off when a window opens? That's JS.
 - A form that checks if you typed your email correctly before sending? That's JS.
- **Key Concept:** JavaScript allows you to **listen for events**. When a user clicks, hovers, or types, JavaScript tells the website to "do something" in response.
- **Why it matters:** Without JS, a website is just a static page—like a printed newspaper. With JS, it becomes an **application** (like an online shop or a social media feed).



4. How they work together (The "Pizza" Analogy)

Now that we understand how HTML, CSS, and JavaScript work as individual web languages, how do they work together in practice?

Imagine a website as a **car that needs to be built**, and web developers as **the mechanics who've been tasked with building it**. They'll take the blueprints — which detail how the car should look in its final form — and turn them into a physical and functioning car that can get the driver from A to B.

To carry through their vision, they have three tools in their arsenal:

- HTML (the building blocks of the car)
- CSS (the style, look, and feel of the car)
- JavaScript (the cars engine and smart features)

First, they'll use HTML to construct the car's basic structure or the 'skeleton'. They'll create door panels, wheels, trunks, and hoods that can be welded together. Not very aesthetically pleasing or functional, but HTML will get the car components built.



Next, CSS is used to embellish the car's basic structure with style and function. Colors are added, as well as materials like leather seats, a wooden dashboard, and aluminium rims. Suddenly the car has gone from a bare-bones structure to something you'd actually want to drive.



Now it's time to take it from show-car to fully functioning vehicle. Using JavaScript, the developers are able to inject some much-needed functionality into the car with a powerful engine, lighting, speakers, dash, and displays. With JavaScript, the car's finishing touches can be added — and before you know it, it's ready to take to the roads with its sleek new look



